The row-sum polynomials of Barry's almost Riordan array

Principia Math in Progress

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Task 1. Prove the row-sum formula. Conjecture 27 of Barry [1] concerns the array produced by $P = \overline{A}$, where the bar deletes the first row and

$$A = \left(\frac{1+2z}{1-z^2}; \frac{-1}{(1-z)(1+z)^2}, \frac{z}{1+z} \right).$$

In this almost Riordan notation the first column has generating function $a(z) = (1+2z)/(1-z^2)$, while column $k \geq 1$ has generating function

$$z \frac{-1}{(1-z)(1+z)^2} \left(\frac{z}{1+z} \right)^{k-1}.$$

Define the produced rows by $u_0 = (1, 0, \ldots)$ and $u_{n+1} = u_n P$, and set $P_n(x) = \sum_k (u_n)_k x^k$. These operations are finite in each row because P is lower Hessenberg.

Theorem 1. *For every integer $n \geq 0$,*

$$P_n(x) = 1 + (1-x) \sum_{i=0}^{n-1} (1-x)^{\overline{i}}, \qquad y^{\overline{i}} = \prod_{j=0}^{i-1} (y+j), \quad y^{\overline{0}} = 1. \tag{1}$$

Factorial convention. The rising factorial in (1) is essential: it gives the source's $P_3 = 5 - 8x + 5x^2 - x^3$. The falling-factorial gloss added in the MathDB transcription would give $3 - 4x + 3x^2 - x^3$, which already contradicts the source's displayed values. We prove the full source formula, rather than counting that transcription error as a counterexample.

Proof. Let $R_m(x) = \sum_k A_{m,k} x^k$. The specified column generating functions give, as a formal power series in z,

$$\begin{aligned} \sum_{m \geq 0} R_m(x) z^m &= \frac{1+2z}{1-z^2} - \frac{zx}{(1-z)(1+z)(1+(1-x)z)} \\ &= \frac{1+2(1-x)z}{(1-z)(1+(1-x)z)} \\ &= \frac{1}{1-z} + \frac{(1-x)z}{(1-z)(1-(x-1)z)}. \end{aligned}$$

Consequently

$$R_{j+1}(x) = 1 + (1-x) \sum_{i=0}^{j} (x-1)^i.$$

Since row j of P is row $j+1$ of A, right multiplication by P acts on row polynomials by the linear operator T defined by $T(x^j) = R_{j+1}(x)$. For every polynomial F, summing the finite geometric series therefore yields

$$(TF)(x) = \frac{(3-2x)F(1) + (1-x)^2 F(x-1)}{2-x}. \tag{2}$$

The numerator vanishes at $x = 2$, so this is a polynomial identity with no excluded value of x. In particular, $P_{n+1} = TP_n$.

Write Q_n for the right side of (1). We have $Q_0 = 1 = P_0$ and $Q_n(1) = 1$. Substitution into (2) gives

$$\begin{aligned} TQ_n &= \frac{3 - 2x + (1-x)^2}{2 - x} + (1-x)^2 \sum_{i=0}^{n-1} (2-x)^{\overline{i}} \\ &= 2 - x + (1-x)^2 \sum_{i=0}^{n-1} (2-x)^{\overline{i}} \\ &= 1 + (1-x)\left(1 + \sum_{i=1}^{n} (1-x)^{\overline{i}}\right) = Q_{n+1}. \end{aligned}$$

The last step uses $(1-x)^{\overline{i+1}} = (1-x)(2-x)^{\overline{i}}$. Induction proves the formula for every n. □

Task 2. Determine an explicit coefficient formula. Express each coefficient of $P_n(x)$ through standard Stirling numbers and obtain a combinatorial interpretation of its absolute value.

Task 3. Analyze the zeros. Determine which P_n have only real zeros, and describe the limiting zero distribution after an appropriate rescaling.

Task 4. Extend the production operator. Identify parameter families of almost Riordan arrays for which the induced operator has an analogous evaluation-and-shift formula and the produced polynomials admit finite factorial sums.

Task 5. Derive asymptotics. Obtain asymptotic formulas for $P_n(x)$ at fixed real or complex x, including the exceptional values where the factorial sum terminates or cancels.

References

[1] P. Barry, *On the partial sums of Riordan arrays*, arXiv:2108.13537v1 (2021). The almost Riordan array is on pp. 19–20, the produced array and its initial polynomials on p. 21, and Conjecture 27 on p. 22 of the inspected 23-page PDF.